

National University of Computer and Emerging Sciences

Lab Manual

Data Analysis and Visualization



Lab 08

Instructor

Rida Mahmood

Class

BDS-5A/5D

Semester

Fall 2023

Fast School of Computing

FAST-NU, Lahore, Paki

References:

Lecture 15 and Lecture 16 of Theory ANN and CNN Concepts

TASK 1

Implementation of ANN from Scratch

Artificial Neural Network

Importing the libraries

import numpy as np

import pandas as pd

import tensorflow as tf

tf.__version__

Part 1 - Data Preprocessing

Importing the dataset

Encoding categorical data

Label Encoding the "Gender" column

One Hot Encoding the "Geography" column

Splitting the dataset into the Training set and Test set

Feature Scaling

Part 2 - Building the ANN

Initializing the ANN and setting its parameters

Initializing and Adding the input layer and the first hidden layer

Adding the second hidden layer if necessary

Adding the output layer

Part 3 - Training the ANN

Compiling the ANN

Training the ANN on the Training set

Part 4 - Making the predictions and evaluating the model

Predicting the result of a single observation

Use our ANN model to predict if the customer with the following informations will leave the bank:

Geography: France
Credit Score: 600
Gender: Male
Age: 40 years old
Tenure: 3 years
Balance: \$ 60000
Number of Products: 2
Does this customer have a credit card? Yes
Is this customer an Active Member: Yes
Estimated Salary: \$ 50000

So, should we say goodbye to that customer?

Therefore, our ANN model predicts that this customer stays in the bank!

Important note 1: Notice that the values of the features were all input in a double pair of square brackets. That's because the "predict" method always expects a 2D array as the format of its inputs. And putting our values into a double pair of square brackets makes the input exactly a 2D array.

Important note 2: Notice also that the "France" country was not input as a string in the last column but as "1, 0, 0" in the first three columns. That's because of course the predict method expects the one-hot-encoded values of the state, and as we see in the first row of the matrix of features X, "France" was encoded as "1, 0, 0". And be careful to include these values in the first three columns, because the dummy variables are always created in the first columns.

Predicting the Test set results
Making the Confusion Matrix

TASK 2

Implementation of CNN from Scratch

```
# Convolutional Neural Network
```

```
# Importing the libraries
```

```
# Part 1 - Data Preprocessing
```

```
# Preprocessing the Training set
```

```
# Preprocessing the Test set
```

```
# Part 2 - Building the CNN
```

```
# Initializing and Building the CNN
```

```
# Step 1 - Convolution
```

```
# Step 2 - Pooling
```

```
# Adding a second convolutional layer
```

```
# Step 3 - Flattening
```

```
# Step 4 - Full Connection
```

```
# Step 5 - Output Layer
```

```
# Part 3 - Training the CNN
```

```
# Compiling the CNN
```

```
# Training the CNN on the Training set and evaluating it on the Test set
```

```
# Part 4 - Making a single prediction
```

```
import numpy as np
```

```
from keras.preprocessing import image
```

```
test_image = image.load_img('dataset/single_prediction/cat_or_dog_1.jpg', target_size = (64, 64))
```

```
test_image = image.img_to_array(test_image)
```

```
test_image = np.expand_dims(test_image, axis = 0)
```

```
result = cnn.predict(test_image)
```

```
training_set.class_indices
```

```
if result[0][0] == 1:
```

```
    prediction = 'dog'
```

```
else:
```

```
    prediction = 'cat'
```

```
print(prediction)
```